

1. Course Title	Applied Forecasting Methods
2. Credit Structure	Lecture hours per week: 3 Tutorial hours per week: 0 Practical hours per week: 2 Total Credits: 4
3. Course Code	SC475
4. Program/Semester	B.Tech. (ICT) (VI and VIII Semester) / M.Tech. (ML) (II Semester) Winter Semester
5. Category	Technical Elective
6. Prerequisite courses	Machine Learning, Basic Programming.
7. Foundation for	UG (VI Semester) / Industry Applications that involves forecasting tasks
8. Abstract Content	This course aims to develop a systematic understanding of the forecasting models from machine learning perspective. The course offers a smooth journey from basic Least Square Regression model developed around year 1805 to modern complex forecasting methods after 2020. The content of the course covers the variants of point and probabilistic forecasting methods with detailed statistical and mathematical analysis for panel and time-series data.
Text Book and Materials	(i) Hyndman, R.J., & Athanasopoulos, G. (2018) Forecasting: principles and practice, 2nd edition, OTexts: Melbourne, Australia. OTexts.com/fpp2 . (ii) Zhang A ,Lipton ZC, Li M and Smola AJ , (2023). <i>Dive into deep learning</i>. Cambridge University Press. (iii) Koenker, R., Chernozhukov, V., He, X., & Peng, L. (Eds.). (2017). Handbook of Quantile Regression (1st ed.). Chapman and Hall/CRC. https://doi.org/10.1201/9781315120256 . (iv) Takeuchi, I., Le, Q., Sears, T., & Smola, A. (2006). Nonparametric quantile estimation, 2006, Journal of Machine Learning and Research , vol. 7, pp. 1231–1264.

Detailed Course Contents		
Topic Name	Content (2 -3 lines per 4 – 6 lectures)	No. of lectures
Introduction to Forecasting Methods	Application of forecasting in different verticals like energy, finance, environment and health. Motivation, Definition, objectives and approaches in Forecasting,	2
Regression from ML perspective	Linear regression models, Least Square Methods, Non-linear regression models, Kernel Regression, Gradient Descent Methods, Evaluation Metrics.	6
Quantile Regression & Uncertainty Quantification	Quantile Regression model, Pinball loss function, Quantile Regression with Kernel Machine, Quantile Regression Neural Network, Prediction Interval Estimation, Evaluation Metrics, Uncertainty Modeling with Kernel Machine and Neural Network.	12
Statistical Time-series Models	Introduction to Time Series Forecasting , AR, ARMA and ARIMA models.	8
Time Series Forecasting with Deep Learning	Time Series Forecasting with Recurrent Neural Network (RNN), Gated Recurrent Unit (GRU), Long Short Term Memory (LSTM) , Deep Auto Regressive (Deep AR) model, Time GPT model.	8
Probabilistic Forecasting	Probabilistic Forecasting in Neural Network, RNN, LSTM and Deep AR models.	4
Outcomes and Objectives	- Systematically develop a deep understanding of the main principles of forecasting. - Develop the machine learning perspective for forecasting and uncertainty modeling. - Develop the understanding of statistical and modern time series models for point and probabilistic forecasting tasks. - Develop the implementation and practical skills for modern and state-of art forecasting methods used in research and industry after COVID 19. - Hands on experience for different forecasting tasks in Finance, Energy, and Sustainability verticals.	
Comments	Lab – The Lab classes would be dedicated for systematic implementation of algorithms for forecasting tasks in Finance, Renewable Energy, Electricity Load, Environment and sustainability. Use cases for Lab class: - • Benchmark Datasets. • Electrical Load Datasets. • Renewable Energy (Wind and Wave) Datasets • Financial and Stock Prices Datasets. • Environment and Pollution Datasets.	

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X			X	X							

Evaluation: -

Written Exam: - 40 %. Assignments: - 20 %, Project +Viva - 40 %